WORKED SOLUTION

STEP III 1999, Question 14

Probability and Statistics • Expectation by Conditioning

STEP III 1999, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q14

The key idea

In H1 the winnings depend only on when the first success happens, so the expectation is a finite arithmetico-geometric sum $\sum r\,p\,q^{r-1}$. The standard trick for such sums is to subtract q times the sum from itself. In H2 we condition on the number of turns N and average the H1 answer over the Poisson distribution. Doing this needs only the Poisson sums $\sum \lambda^n x^n/n!$ and $\sum n\lambda^n x^n/n!$, both of which come from the exponential series.

Solution

Game H1

Let W be her winnings in pounds. Turns are independent, each a success with probability p. Her first success is at turn r exactly when the first $r-1$ turns fail and turn r succeeds, so

$$\mathrm{P}(W=r) = q^{r-1}p \qquad (1 \le r \le n),$$

and otherwise $W=0$, which adds nothing to the expectation. Hence

$$\mathrm{E}(W) = p\,S, \qquad \text{where } S = \sum_{r=1}^{n} r\,q^{r-1} = 1 + 2q + 3q^2 + \cdots + nq^{n-1}.$$

Multiply by q and subtract, lining up equal powers of q:

$$\begin{aligned} S - qS &= 1 + q + q^2 + \cdots + q^{n-1} - nq^n \\ &= \frac{1-q^n}{1-q} - nq^n, \end{aligned}$$

by the formula for a geometric series, which is valid because $q \neq 1$. Since $1 - q = p$, this gives $S = \dfrac{1-q^n}{p^2} - \dfrac{nq^n}{p}$, and so

$$\begin{aligned} \mathrm{E}(W) = pS &= \frac{1 - q^n - npq^n}{p} = \frac{1 - q^n - n(1-q)q^n}{p} \\ &= p^{-1}\left[1 + nq^{n+1} - (n+1)q^n\right], \end{aligned}$$

which is the required result.

Game H2

Let $N \sim \mathrm{Po}(\lambda)$ be the number of turns allowed, so $\mathrm{P}(N = n) = \mathrm{e}^{-\lambda}\lambda^n/n!$. Write

$$g(n) = p^{-1}\left[1 + nq^{n+1} - (n+1)q^n\right].$$

Given $N = n \geq 1$, she plays H1 with n turns, so her expected winnings are $g(n)$. Given $N = 0$ she wins nothing. By the law of total expectation (conditioning on N),

$$\mathrm{E}(W) = \sum_{n=1}^{\infty} \mathrm{P}(N = n)\, g(n).$$

Ask yourself The three pieces of $g(n)$ would be easier to handle as full Poisson sums from $n = 0$. Can we start the sum at $n = 0$?

Yes, we can: $g(0) = p^{-1}[1 + 0 - 1] = 0$, which agrees with the winnings of nil when $N = 0$. So

$$\mathrm{E}(W) = \frac{1}{p}\sum_{n=0}^{\infty} \mathrm{e}^{-\lambda}\frac{\lambda^n}{n!}\left[1 + nq^{n+1} - nq^n - q^n\right].$$

The first term sums to 1, since the probabilities add to 1. For the others we need two series, both valid for every real x:

$$\begin{aligned} \sum_{n=0}^{\infty} \mathrm{e}^{-\lambda}\frac{(\lambda x)^n}{n!} &= \mathrm{e}^{-\lambda}\mathrm{e}^{\lambda x}, \\ \sum_{n=0}^{\infty} n\,\mathrm{e}^{-\lambda}\frac{(\lambda x)^n}{n!} &= \lambda x \sum_{n=1}^{\infty} \mathrm{e}^{-\lambda}\frac{(\lambda x)^{n-1}}{(n-1)!} = \lambda x\,\mathrm{e}^{-\lambda}\mathrm{e}^{\lambda x}. \end{aligned}$$

Put $x = q$, and note that $\mathrm{e}^{-\lambda}\mathrm{e}^{\lambda q} = \mathrm{e}^{-\lambda(1-q)} = \mathrm{e}^{-\lambda p}$. Then

$$\sum_{n=0}^{\infty} \mathrm{e}^{-\lambda}\frac{\lambda^n}{n!}\, q^n = \mathrm{e}^{-\lambda p}, \qquad \sum_{n=0}^{\infty} \mathrm{e}^{-\lambda}\frac{\lambda^n}{n!}\, nq^n = \lambda q\,\mathrm{e}^{-\lambda p}.$$

The term nq^{n+1} is q times nq^n, so it contributes $\lambda q^2 \mathrm{e}^{-\lambda p}$. Altogether,

$$\begin{aligned} \mathrm{E}(W) &= \frac{1}{p}\left[1 + \lambda q^2 \mathrm{e}^{-\lambda p} - \lambda q\, \mathrm{e}^{-\lambda p} - \mathrm{e}^{-\lambda p}\right] \\ &= \frac{1}{p}(1 - \mathrm{e}^{-\lambda p}) + \frac{\lambda q(q-1)}{p}\, \mathrm{e}^{-\lambda p} \\ &= \frac{1}{p}(1 - \mathrm{e}^{-\lambda p}) - \lambda q\, \mathrm{e}^{-\lambda p}, \end{aligned}$$

using $q - 1 = -p$ in the last step. This is the required expression, in pounds.

Checks

- H1 with $n = 1$: the formula gives $p^{-1}[1 + q^2 - 2q] = p^{-1}(1-q)^2 = p$. This is right, because she wins £1 with probability p.
- As $n \to \infty$ in H1, $q^n \to 0$ and $nq^n \to 0$, so the expectation tends to $1/p$, the mean of a geometric distribution. Similarly, H2 tends to $1/p$ as $\lambda \to \infty$.
- For small λ, expanding the exponentials gives $\lambda - \frac{1}{2}\lambda^2 p - \lambda q + O(\lambda^2) = \lambda p + O(\lambda^2)$. This makes sense: she almost always gets either no turns or one turn, and $\mathrm{P}(N = 1) \approx \lambda$.
- Numerically, the H1 formula matched the direct sum exactly (using fractions) for $n \leq 14$. The H2 formula matched the truncated Poisson sum, and a simulation of 4×10^5 games with $\lambda = 2$, $p = 0.4$ gave 0.836, against the formula's 0.837.

Common slips

- Getting the last term of $S - qS$ wrong: the sum S stops at nq^{n-1}, so qS leaves $-nq^n$ over.
- Forgetting that $\mathrm{e}^{-\lambda}\mathrm{e}^{\lambda q}$ simplifies to $\mathrm{e}^{-\lambda p}$, and then getting lost in the algebra.

Beyond STEP

The sums $\sum \mathrm{P}(N = n)\, x^n$ used in H2 are values of the *probability generating function* $G_N(x) = \mathrm{E}(x^N)$, and $\sum n\, \mathrm{P}(N = n)\, x^n = xG'_N(x)$. For a Poisson variable, $G_N(x) = \mathrm{e}^{\lambda(x-1)}$. Generating functions, together with conditioning on a random number of trials, are central tools in first-year probability, for example in the study of random sums and branching processes. Grimmett and Welsh, *Probability: An Introduction*, covers both clearly.

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